

JNTU Hyderabad

MID-TERM EXAMINATION ? 2025

Engineering Mechanics ? Mid-Term Examination, 2025

Subject Code: ME105 Semester: 2 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Statics of Particles)

1. Write short notes on Statics of Particles. [12 Marks]
2. List the advantages and limitations of Statics of Particles. Explain any two in detail. [12 Marks]
3. List the advantages and limitations of Statics of Particles. Explain any two in detail. [12 Marks]

SECTION B (Unit 2: Equilibrium of Rigid Bodies)

4. Write short notes on Equilibrium of Rigid Bodies. [12 Marks]
5. Discuss the engineering applications of Equilibrium of Rigid Bodies. [8 Marks]
6. Derive the expression related to Equilibrium of Rigid Bodies and explain each term. [8 Marks]

SECTION C (Unit 3: Centroid and Moment of Inertia)

7. Compare and contrast the key concepts involved in Centroid and Moment of Inertia. [12 Marks]
8. Derive the expression related to Centroid and Moment of Inertia and explain each term. [8 Marks]
9. How is Centroid and Moment of Inertia implemented? Give an algorithm / procedure and explain. [8 Marks]

SECTION D (Unit 4: Dynamics of Particles)

10. Define Dynamics of Particles. Explain its significance in Engineering Mechanics with a suitable example. [12 Marks]
11. Solve a problem related to Dynamics of Particles. Show all the steps clearly. [12 Marks]
12. Derive the expression related to Dynamics of Particles and explain each term. [12 Marks]

SECTION E (Unit 5: Friction and Virtual Work)

13. Derive the expression related to Friction and Virtual Work and explain each term. [12 Marks]
14. How is Friction and Virtual Work implemented? Give an algorithm / procedure and

explain. [12 Marks]

15. Describe the working principle of Friction and Virtual Work with necessary equations. [12 Marks]

*** End of Question Paper ***